insight. And for this purpose, Hawking believed, the new, absolute horizon, with its smoothly increasing area, would be superior to the old, apparent horizon, with its discontinuous jumps in size.

Stephen Hawking was not the first physicist to think about absolute horizons and discover their area increase. Roger Penrose at Oxford had already done so, before Hawking's sleepless November night. In fact, Hawking's insights were based largely on foundations laid by Penrose (Chapter 13) and a recent conversation with him. However, Penrose had not recognized the power of the area-increase theorem, and so had not perfected nor published it. Why had he missed its power? Because he had not developed any clear mental picture of the absolute horizon's location. He had missed what Hawking saw so clearly that sleepless night: After black holes collide, their merged absolute horizon should soon settle down into a quiescent location whose surface area can be computed from the standard equations for quiescent black holes.

Werner Israel at the University of Alberta, Canada, also caught a glimpse of the area increase theorem before Hawking; but, failing to recognize its importance, he, too, let it slip by unpublished. Moreover, by contrast with Hawking and Penrose, Israel was still using—indeed, was hypnotized by—the old concept of the apparent horizon, as were all the rest of us relativity theorists. The apparent horizon had played a central role in Penrose's amazing 1964 discovery that Einstein's laws force every black hole to have a singularity at its center (Chapter 13). The apparent horizon thereby had proved its power. Blinded by that power, we relativists could not conceive of replacing the apparent horizon, as the definition of a black hole's surface, by the absolute horizon.

We also were paying little attention to the absolute horizon because it violates our cherished notion that an effect should not precede its cause. When matter falls into a black hole, the absolute horizon starts to grow ("effect") before the matter reaches it ("cause"). The horizon grows in anticipation that the matter will soon be swallowed and will increase the hole's gravitational pull (Box 12.2). This seeming paradox has a simple origin. The very definition of the absolute horizon depends on what will happen in the future: on whether or not signals will ultimately escape to the distant Universe. In the terminology of philosophers, it is a *teleological* definition (a definition that relies on "final causes"), and it forces the